Configuração 1: $M = 128$, $C = 8$, $K = 8$, IoT = 10 dB, 16-QAM

Perda de SNR do BCD-MMSE (iter 1) em SER = 10^{-2}, em dB, em relação à referência ideal (float64); × = não chega a 10^{-2} entre -10 e 10 dB

TDLA30-75: $N = 192$, $f_c = 28$ GHz

	I=6	I=8	I=10	I=12	I=14	I=16	I=20	I=24	I=32
AP_TRN / AP_SAT	2.3	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
AP_TRN / AP_SAT_ZERO	2.7	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
AP_TRN / AP_SAT_SYM	2.3	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
AP_TRN / AP_WRAP	1.5	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
AP_TRN / AP_WRAP_SM	2.7	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
AP_TRN_ZERO / AP_SAT	2.3	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
AP_TRN_ZERO / AP_SAT_ZERO	2.7	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
AP_TRN_ZERO / AP_SAT_SYM	2.3	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
AP_TRN_ZERO / AP_WRAP	1.5	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
AP_TRN_ZERO / AP_WRAP_SM	2.7	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
AP_RND / AP_SAT	2.3	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
AP_RND / AP_SAT_ZERO	2.7	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
AP_RND / AP_SAT_SYM	2.3	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
AP_RND / AP_WRAP	1.5	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
AP_RND / AP_WRAP_SM	2.7	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
AP_RND_ZERO / AP_SAT	2.3	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
AP_RND_ZERO / AP_SAT_ZERO	2.7	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
AP_RND_ZERO / AP_SAT_SYM	2.3	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
AP_RND_ZERO / AP_WRAP	1.5	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
AP_RND_ZERO / AP_WRAP_SM	2.7	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
AP_RND_MIN_INF / AP_SAT	2.3	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
AP_RND_MIN_INF / AP_SAT_ZERO	2.7	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
AP_RND_MIN_INF / AP_SAT_SYM	2.3	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
AP_RND_MIN_INF / AP_WRAP	1.5	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
AP_RND_MIN_INF / AP_WRAP_SM	2.7	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
AP_RND_INF / AP_SAT	2.3	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
AP_RND_INF / AP_SAT_ZERO	2.7	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
AP_RND_INF / AP_SAT_SYM	2.3	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
AP_RND_INF / AP_WRAP	1.5	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
AP_RND_INF / AP_WRAP_SM	2.7	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
AP_RND_CONV / AP_SAT	2.3	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
AP_RND_CONV / AP_SAT_ZERO	2.7	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
AP_RND_CONV / AP_SAT_SYM	2.3	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
AP_RND_CONV / AP_WRAP	1.5	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
AP_RND_CONV / AP_WRAP_SM	2.7	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0

Bits inteiros I (F = 32)

TDLA30-75: $N = 192$, $f_c = 28$ GHz

	F=4	F=6	F=8	F=10	F=12	F=16	F=20	F=24	F=32
AP_TRN / AP_SAT	×	×	3.4	0.2	0.0	0.0	0.0	0.0	0.0
AP_TRN / AP_SAT_ZERO	×	×	3.4	0.2	0.0	0.0	0.0	0.0	0.0
AP_TRN / AP_SAT_SYM	×	×	3.4	0.2	0.0	0.0	0.0	0.0	0.0
AP_TRN / AP_WRAP	×	×	3.4	0.2	0.0	0.0	0.0	0.0	0.0
AP_TRN / AP_WRAP_SM	×	×	3.4	0.2	0.0	0.0	0.0	0.0	0.0
AP_TRN_ZERO / AP_SAT	×	×	1.0	0.0	0.0	0.0	0.0	0.0	0.0
AP_TRN_ZERO / AP_SAT_ZERO	×	×	1.0	0.0	0.0	0.0	0.0	0.0	0.0
AP_TRN_ZERO / AP_SAT_SYM	×	×	1.0	0.0	0.0	0.0	0.0	0.0	0.0
AP_TRN_ZERO / AP_WRAP	×	×	1.0	0.0	0.0	0.0	0.0	0.0	0.0
AP_TRN_ZERO / AP_WRAP_SM	×	×	1.0	0.0	0.0	0.0	0.0	0.0	0.0
AP_RND / AP_SAT	×	6.5	0.5	0.0	0.0	0.0	0.0	0.0	0.0
AP_RND / AP_SAT_ZERO	×	6.5	0.5	0.0	0.0	0.0	0.0	0.0	0.0
AP_RND / AP_SAT_SYM	×	6.5	0.5	0.0	0.0	0.0	0.0	0.0	0.0
AP_RND / AP_WRAP	×	6.5	0.5	0.0	0.0	0.0	0.0	0.0	0.0
AP_RND / AP_WRAP_SM	×	6.5	0.5	0.0	0.0	0.0	0.0	0.0	0.0
AP_RND_ZERO / AP_SAT	×	6.4	0.5	0.0	0.0	0.0	0.0	0.0	0.0
AP_RND_ZERO / AP_SAT_ZERO	×	6.4	0.5	0.0	0.0	0.0	0.0	0.0	0.0
AP_RND_ZERO / AP_SAT_SYM	×	6.4	0.5	0.0	0.0	0.0	0.0	0.0	0.0
AP_RND_ZERO / AP_WRAP	×	6.4	0.5	0.0	0.0	0.0	0.0	0.0	0.0
AP_RND_ZERO / AP_WRAP_SM	×	6.4	0.5	0.0	0.0	0.0	0.0	0.0	0.0
AP_RND_MIN_INF / AP_SAT	×	6.6	0.5	0.0	0.0	0.0	0.0	0.0	0.0
AP_RND_MIN_INF / AP_SAT_ZERO	×	6.6	0.5	0.0	0.0	0.0	0.0	0.0	0.0
AP_RND_MIN_INF / AP_SAT_SYM	×	6.6	0.5	0.0	0.0	0.0	0.0	0.0	0.0
AP_RND_MIN_INF / AP_WRAP	×	6.6	0.5	0.0	0.0	0.0	0.0	0.0	0.0
AP_RND_MIN_INF / AP_WRAP_SM	×	6.6	0.5	0.0	0.0	0.0	0.0	0.0	0.0
AP_RND_INF / AP_SAT	×	6.6	0.5	0.0	0.0	0.0	0.0	0.0	0.0
AP_RND_INF / AP_SAT_ZERO	×	6.6	0.5	0.0	0.0	0.0	0.0	0.0	0.0
AP_RND_INF / AP_SAT_SYM	×	6.6	0.5	0.0	0.0	0.0	0.0	0.0	0.0
AP_RND_INF / AP_WRAP	×	6.6	0.5	0.0	0.0	0.0	0.0	0.0	0.0
AP_RND_INF / AP_WRAP_SM	×	6.6	0.5	0.0	0.0	0.0	0.0	0.0	0.0
AP_RND_CONV / AP_SAT	×	6.2	0.5	0.0	0.0	0.0	0.0	0.0	0.0
AP_RND_CONV / AP_SAT_ZERO	×	6.2	0.5	0.0	0.0	0.0	0.0	0.0	0.0
AP_RND_CONV / AP_SAT_SYM	×	6.2	0.5	0.0	0.0	0.0	0.0	0.0	0.0
AP_RND_CONV / AP_WRAP	×	6.2	0.5	0.0	0.0	0.0	0.0	0.0	0.0
AP_RND_CONV / AP_WRAP_SM	×	6.2	0.5	0.0	0.0	0.0	0.0	0.0	0.0

Bits fracionários F (I = 14)

TDLB100-400: $N = 24$, $f_c = 3{,}5$ GHz

	I=6	I=8	I=10	I=12	I=14	I=16	I=20	I=24	I=32
AP_TRN / AP_SAT	0.9	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
AP_TRN / AP_SAT_ZERO	2.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
AP_TRN / AP_SAT_SYM	0.9	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
AP_TRN / AP_WRAP	0.9	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
AP_TRN / AP_WRAP_SM	1.9	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
AP_TRN_ZERO / AP_SAT	0.9	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
AP_TRN_ZERO / AP_SAT_ZERO	2.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
AP_TRN_ZERO / AP_SAT_SYM	0.9	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
AP_TRN_ZERO / AP_WRAP	0.9	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
AP_TRN_ZERO / AP_WRAP_SM	1.9	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
AP_RND / AP_SAT	0.9	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
AP_RND / AP_SAT_ZERO	2.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
AP_RND / AP_SAT_SYM	0.9	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
AP_RND / AP_WRAP	0.9	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
AP_RND / AP_WRAP_SM	1.9	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
AP_RND_ZERO / AP_SAT	0.9	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
AP_RND_ZERO / AP_SAT_ZERO	2.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
AP_RND_ZERO / AP_SAT_SYM	0.9	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
AP_RND_ZERO / AP_WRAP	0.9	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
AP_RND_ZERO / AP_WRAP_SM	1.9	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
AP_RND_MIN_INF / AP_SAT	0.9	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
AP_RND_MIN_INF / AP_SAT_ZERO	2.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
AP_RND_MIN_INF / AP_SAT_SYM	0.9	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
AP_RND_MIN_INF / AP_WRAP	0.9	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
AP_RND_MIN_INF / AP_WRAP_SM	1.9	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
AP_RND_INF / AP_SAT	0.9	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
AP_RND_INF / AP_SAT_ZERO	2.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
AP_RND_INF / AP_SAT_SYM	0.9	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
AP_RND_INF / AP_WRAP	0.9	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
AP_RND_INF / AP_WRAP_SM	1.9	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
AP_RND_CONV / AP_SAT	0.9	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
AP_RND_CONV / AP_SAT_ZERO	2.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
AP_RND_CONV / AP_SAT_SYM	0.9	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
AP_RND_CONV / AP_WRAP	0.9	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
AP_RND_CONV / AP_WRAP_SM	1.9	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0

Bits inteiros I (F = 32)

TDLB100-400: $N = 24$, $f_c = 3{,}5$ GHz

	F=4	F=6	F=8	F=10	F=12	F=16	F=20	F=24	F=32
AP_TRN / AP_SAT	×	×	2.4	0.1	0.0	0.0	0.0	0.0	0.0
AP_TRN / AP_SAT_ZERO	×	×	2.4	0.1	0.0	0.0	0.0	0.0	0.0
AP_TRN / AP_SAT_SYM	×	×	2.4	0.1	0.0	0.0	0.0	0.0	0.0
AP_TRN / AP_WRAP	×	×	2.4	0.1	0.0	0.0	0.0	0.0	0.0
AP_TRN / AP_WRAP_SM	×	×	2.4	0.1	0.0	0.0	0.0	0.0	0.0
AP_TRN_ZERO / AP_SAT	×	×	1.5	0.1	0.0	0.0	0.0	0.0	0.0
AP_TRN_ZERO / AP_SAT_ZERO	×	×	1.5	0.1	0.0	0.0	0.0	0.0	0.0
AP_TRN_ZERO / AP_SAT_SYM	×	×	1.5	0.1	0.0	0.0	0.0	0.0	0.0
AP_TRN_ZERO / AP_WRAP	×	×	1.5	0.1	0.0	0.0	0.0	0.0	0.0
AP_TRN_ZERO / AP_WRAP_SM	×	×	1.5	0.1	0.0	0.0	0.0	0.0	0.0
AP_RND / AP_SAT	×	6.0	0.5	0.0	0.0	0.0	0.0	0.0	0.0
AP_RND / AP_SAT_ZERO	×	6.0	0.5	0.0	0.0	0.0	0.0	0.0	0.0
AP_RND / AP_SAT_SYM	×	6.0	0.5	0.0	0.0	0.0	0.0	0.0	0.0
AP_RND / AP_WRAP	×	6.0	0.5	0.0	0.0	0.0	0.0	0.0	0.0
AP_RND / AP_WRAP_SM	×	6.0	0.5	0.0	0.0	0.0	0.0	0.0	0.0
AP_RND_ZERO / AP_SAT	×	5.8	0.5	0.0	0.0	0.0	0.0	0.0	0.0
AP_RND_ZERO / AP_SAT_ZERO	×	5.8	0.5	0.0	0.0	0.0	0.0	0.0	0.0
AP_RND_ZERO / AP_SAT_SYM	×	5.8	0.5	0.0	0.0	0.0	0.0	0.0	0.0
AP_RND_ZERO / AP_WRAP	×	5.8	0.5	0.0	0.0	0.0	0.0	0.0	0.0
AP_RND_ZERO / AP_WRAP_SM	×	5.8	0.5	0.0	0.0	0.0	0.0	0.0	0.0
AP_RND_MIN_INF / AP_SAT	×	6.0	0.5	0.0	0.0	0.0	0.0	0.0	0.0
AP_RND_MIN_INF / AP_SAT_ZERO	×	6.0	0.5	0.0	0.0	0.0	0.0	0.0	0.0
AP_RND_MIN_INF / AP_SAT_SYM	×	6.0	0.5	0.0	0.0	0.0	0.0	0.0	0.0
AP_RND_MIN_INF / AP_WRAP	×	6.0	0.5	0.0	0.0	0.0	0.0	0.0	0.0
AP_RND_MIN_INF / AP_WRAP_SM	×	6.0	0.5	0.0	0.0	0.0	0.0	0.0	0.0
AP_RND_INF / AP_SAT	×	6.0	0.5	0.0	0.0	0.0	0.0	0.0	0.0
AP_RND_INF / AP_SAT_ZERO	×	6.0	0.5	0.0	0.0	0.0	0.0	0.0	0.0
AP_RND_INF / AP_SAT_SYM	×	6.0	0.5	0.0	0.0	0.0	0.0	0.0	0.0
AP_RND_INF / AP_WRAP	×	6.0	0.5	0.0	0.0	0.0	0.0	0.0	0.0
AP_RND_INF / AP_WRAP_SM	×	6.0	0.5	0.0	0.0	0.0	0.0	0.0	0.0
AP_RND_CONV / AP_SAT	×	6.0	0.5	0.0	0.0	0.0	0.0	0.0	0.0
AP_RND_CONV / AP_SAT_ZERO	×	6.0	0.5	0.0	0.0	0.0	0.0	0.0	0.0
AP_RND_CONV / AP_SAT_SYM	×	6.0	0.5	0.0	0.0	0.0	0.0	0.0	0.0
AP_RND_CONV / AP_WRAP	×	6.0	0.5	0.0	0.0	0.0	0.0	0.0	0.0
AP_RND_CONV / AP_WRAP_SM	×	6.0	0.5	0.0	0.0	0.0	0.0	0.0	0.0

Bits fracionários F (I = 14)